

Total No. of Questions : 8]

SEAT No. :

PD4700

[Total No. of Pages : 2

[6404]-206

B.E. (Mechanical Engineering)
Elective - III: AUTOMOBILE DESIGN
(2019 Pattern) (Semester - VII) (402044A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

- Q1) a)** A propeller shaft is required to transmit 45 kW power at 500 rpm. It is a hollow shaft, having an inside diameter 0.6 times of outside diameter. It is made of plain carbon steel and the permissible shear stress is 84 N/mm². Calculate the inside and outside diameters of the shaft. **[9]**
- b)** Explain the construction and working of a rack and pinion type steering gear. **[9]**

OR

- Q2) a)** Differentiate between semi floating and full floating axle. **[8]**
- b)** A pair of cast iron bevel gears connect two shafts at right angles. The pitch diameters of the pinion and gear are 80 mm and 100 mm respectively. The tooth profiles of the gears are of $14\frac{1}{2}^\circ$ composite form. The allowable static stress for both the gears is 55 MPa. If the pinion transmits 2.75 kW at 1100 r.p.m., find the module and number of teeth on each gear from the standpoint of strength and check the design from the standpoint of wear. Take surface endurance limit as 630 MPa and modulus of elasticity for cast iron as 84 kN/mm². **[10]**

- Q3) a)** What is ABS? Explain with neat sketch working of ABS. **[9]**
- b)** What is the angle of banking necessary for a curved road of 50 m radius for safe driving at 54 km/h? If the road is not banked, what is the coefficient of friction necessary between the road surface and tyres for safe driving at this speed? **[8]**

OR

- Q4) a)** Write a short note on **[10]**
- i) Classification of pressed steel disc wheel
 - ii) Factors affecting tyre life
- b)** Compare radial ply tyre over the cross ply tyre. **[7]**

P.T.O.

- Q5) a)** Explain with neat sketch MacPherson strut front independent suspension system. State its advantages. [9]
- b)** A helical compression spring is used to absorb the shock. The initial compression of the spring is 30 mm and it is further compressed by 50 mm while absorbing the shock. The spring is to absorb 250 J of energy during the process. The spring index can be taken as 6. The spring is made of patented and cold-drawn steel wire with an ultimate tensile strength of 1500 N/mm² and modulus of rigidity of 81 370 N/mm². The permissible shear stress for the spring wire should be taken as 30% of the ultimate tensile strength. [9]
- Calculate:
- i) wire diameter;
 - ii) mean coil diameter;
 - iii) number of active turns

OR

- Q6) a)** Explain with neat sketch hydro gas suspension system. State its advantages. [9]
- b)** A semi-elliptic leaf spring used for automobile suspension consists of three extra full-length leaves and 15 graduated-length leaves, including the master leaf. The centre-to-centre distance between two eyes of the spring is 1 m. The maximum force that can act on the spring is 75 kN. For each leaf, the ratio of width to thickness is 9:1. The modulus of elasticity of the leaf material is 207 000 N/mm². The leaves are pre-stressed in such a way that when the force is maximum, the stresses induced in all leaves are same and equal to 450 N/mm². Determine. [9]
- i) the width and thickness of the leaves;
 - ii) the initial nip; and
 - iii) the initial pre-load required to close the gap C between extra full length leaves and graduated-length leaves.

- Q7) a)** Explain the following [10]
- i) Mechanical packaging
 - ii) Automotive industry packaging
- b)** Discuss steps involved in driver package development. [7]

OR

- Q8) a)** Explain with suitable examples anthropometry in automobile. [9]
- b)** Write a note on applications of biomechanics in vehicle. [8]

